Questions by topic: Energy conservation: transport, buildings, industry - Set 2 (15 marks)

Answer all questions in the spaces provided.

0 1 A 1960s office block is being refurbished. Table 1 shows the measured heat loss through its fabric before the work.

Table 1

Part of the building	Heat loss / kW
Walls (uninsulated)	42
Roof	26
Single-glazed windows	31
Floor	9
Draughts and ventilation	22

0 1 . 1 Calculate the percentage of the total heat loss that passes through the windows.

[2 marks]

Answer ☐ %

0 1 . 2 Explain how each of the following reduces heat loss.

[3 marks]

Cavity wall insulation .....

.....

Low-emissivity double glazing .....

.....

A heat exchanger on the ventilation system .....

.....

0 1 . 3 Explain why there is an optimum thickness of loft insulation rather than simply using as much as possible.

[2 marks]

.....

Do not write outside the box

Do not write outside the box

0 1 . 4 The refurbishment also includes a combined heat and power (CHP) unit. Explain why a CHP unit has a lower **electrical** efficiency than a large power station but a much higher **overall** efficiency.

[3 marks]

0 1 . 5 Explain **five** other ways the refurbished building could be designed or managed to use less energy.

[5 marks]

15

END OF QUESTIONS